Prometheus Rocket Final Presentation

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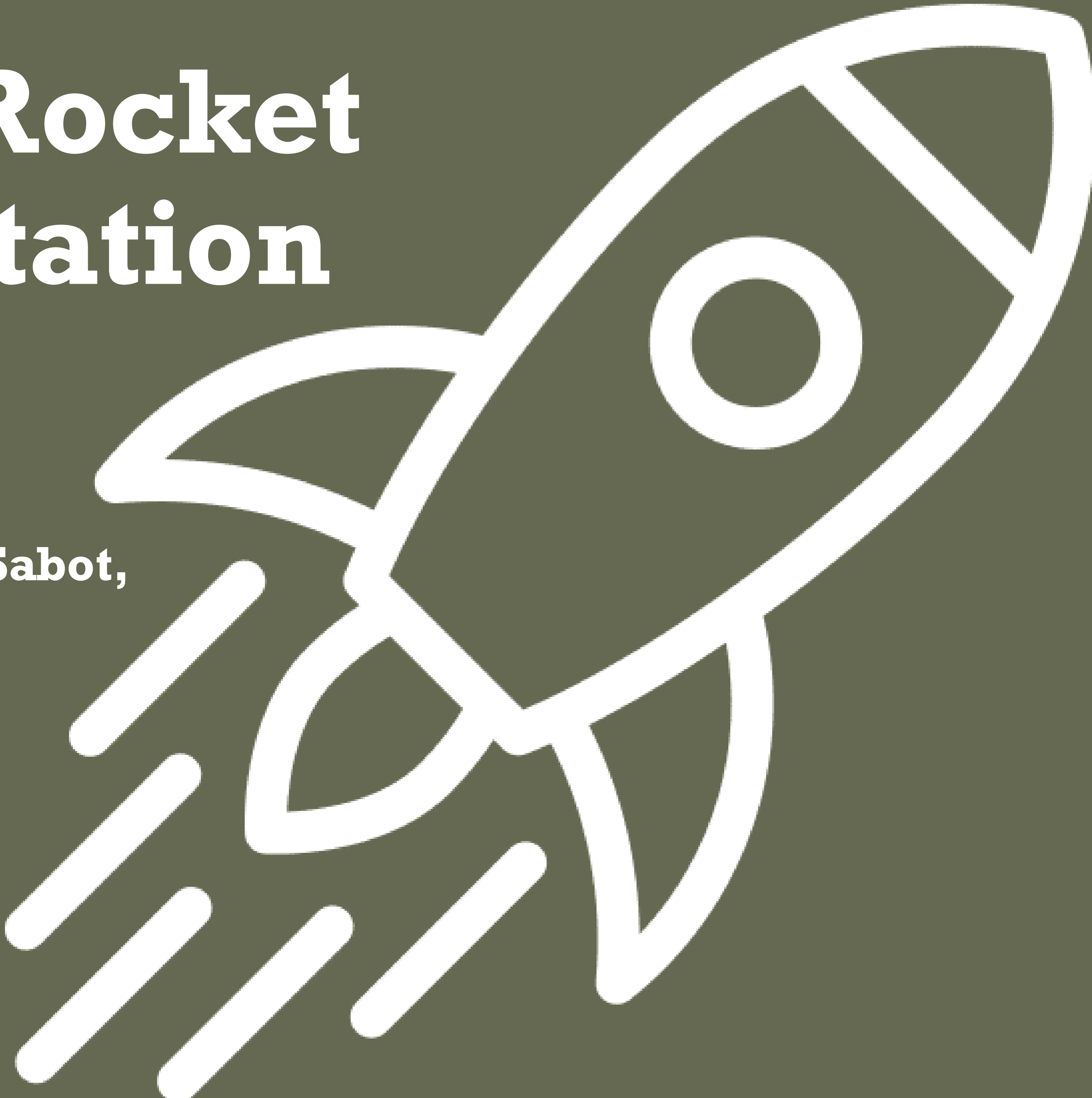
Walla Walla University

SEVENTH-DAY ADVENTIST HIGHER EDUCATION



Previous Projects

Icarus

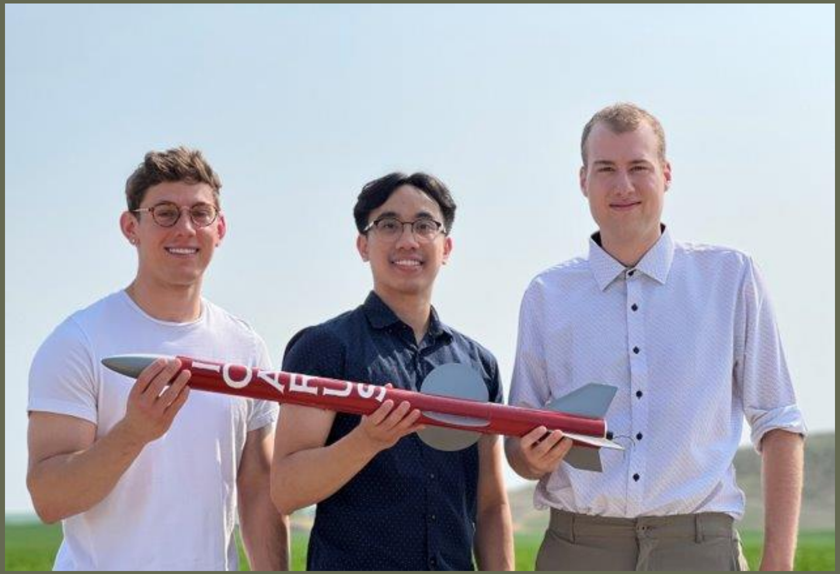


Figure 1. 2023 Icarus Project Team (Icarus, 2024)

- Two-Stage, Cardboard Body Rocket
- Max Altitude: 911 m

Vulcan

Figure 2. 2024 Vulcan Project Team (Vulcan, 2025)

- Two-Stage, Fiberglass Body Rocket
- Max Altitude: 1355 m

Project Objective

- Design a high-powered two stage rocket
- Create an active control system for in-flight stability
- Create a stage separation system
- Implement dual deploy recovery system in both stages
- Design and manufacture a composite tube for both stages

Prometheus

- Max Altitude: ~7ft
- Max Velocity: ~75 mph
- We have not launched

Test Section Testing

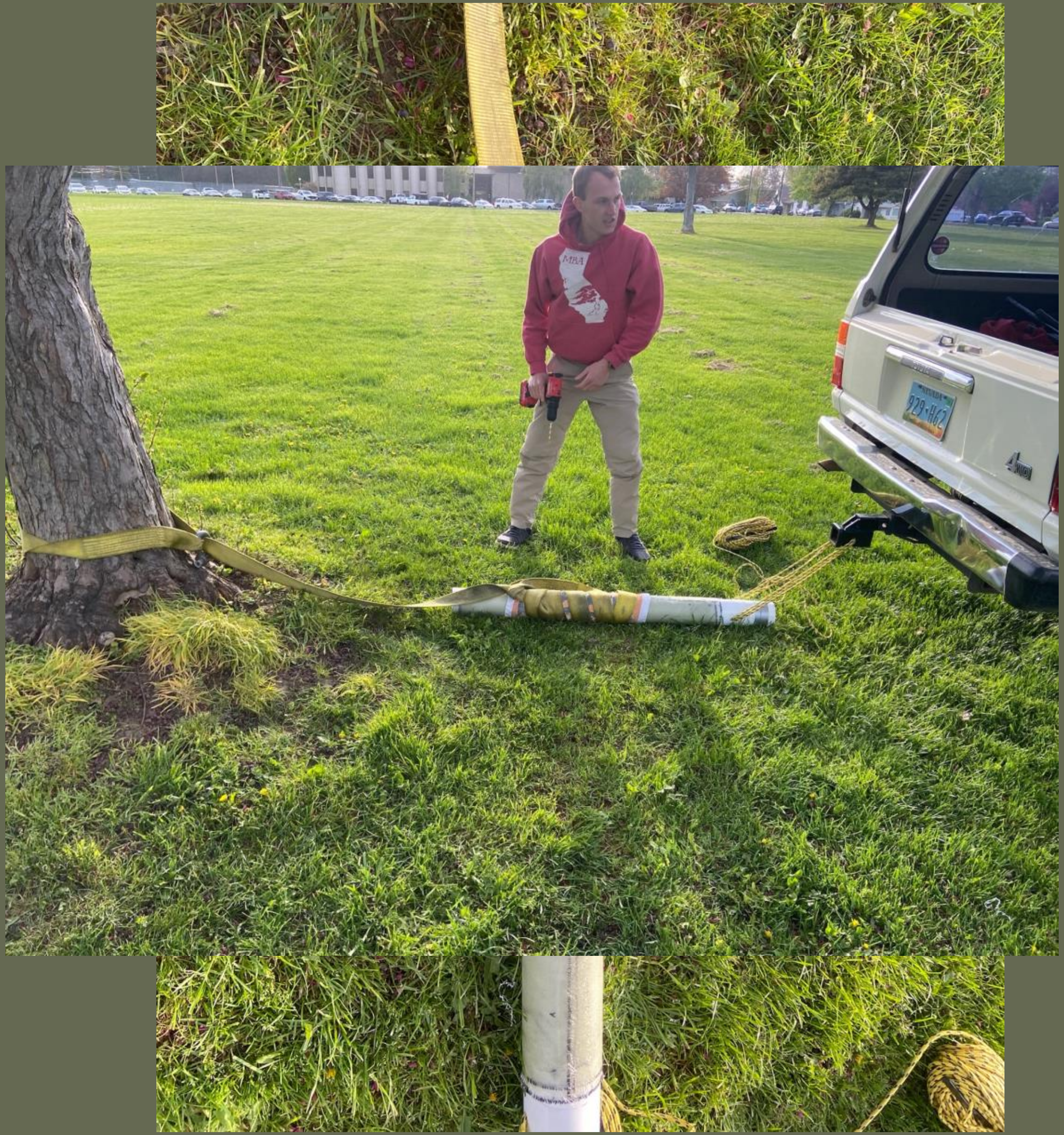
- 6- layer fiberglass-epoxy composite
- 4in long
- Buckled at 2700lb (12,000N) in compression

Tube Manufacturing

- 6-layer fiberglass $[0,45]_3$
- 0.7 fiber volume fraction
- 2 fiberglass layers wrapped 3 times

Tube Removal

- Methods tried:
  - Cut PVC to induce fracture
  - Dremel
  - Thermal (Hot/Cold)
  - Melting the PVC
  - Land Cruiser, Tow Strap



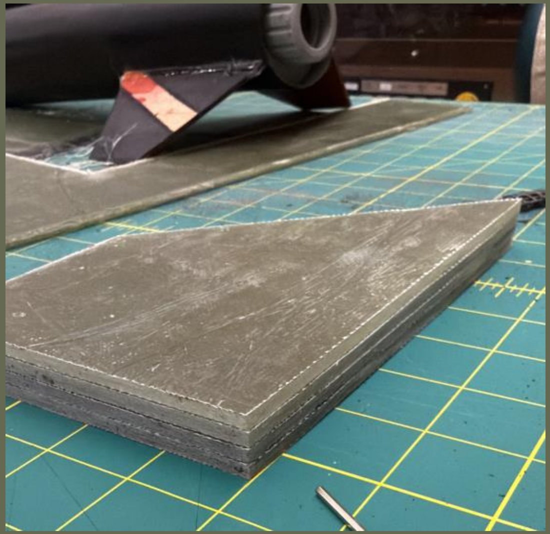

TOYOTA LANDCRUISER
929-H62

First Stage Fins

Manufacturing Method
- Used fin root as the primary datum surface
- Flattened root edge on granite surface plate
- Checked perpendicular reference edge with a square
- Measured remaining geometry from these two references

Why It Mattered
- Improved repeatability between fins
- Reduced angular error during layout
- Helped maintain symmetry and stable flight

Geometry:
Trapezoidal fin set, 4 fins
Root chord: 18 cm
Tip chord: 8 cm
Height: 11 cm
Sweep length: 10 cm



Fin Attachment

Aerodynamics

- Fin Shape Iteration
- Initial idea: symmetric, tapered fin profile
- Looked streamlined for incoming and exiting flow
- OpenRocket stability was too low, about **1.3 cal**
- Swept-back design moved CP rearward and improved stability
- Why Sweep Back Helped

Fuselage Roughness:

$Re_x = \frac{V\,x}{\nu}$ $\qquad \delta = \frac{0.37\,x}{Re_x^{\frac{1}{5}}}$ $\qquad k < \frac{\delta}{10}$

$Re_x = \frac{(343)1.82}{1.5\cdot 10^{-5}} = 4.16 \cdot 10^7$ $\qquad \delta = \frac{0.37\,(1.82)}{(4.16\cdot 10^7)^{\frac{1}{5}}} = 0.0202\,m \rightarrow 20.2\,mm$

$$k < \frac{20.2}{10} \;\rightarrow\; k < 2.02\,mm$$

Fin Roughness:

$x = \frac{0.18\,m + 0.08\,m}{2} = 0.13\,m$ $\qquad Re_x = \frac{V\,x}{\upsilon} = \frac{343(0.13)}{1.5\cdot 10^{-5}} = 2.97 \cdot 10^6$

$\delta = \frac{0.37\,(0.13)}{(2.97\cdot 10^6)^{\frac{1}{5}}} = 0.00245\,m \;\rightarrow\; 2.45\,mm$

$k < \frac{2.45}{10}$ $\qquad k < \frac{2.45}{10} \;\rightarrow\; k < 0.245\text{mm}$

3D Printing considerations

- Stress Concentrations
- Strength Settings
- Infill Choice
- Total Printing Time: 140+ hours

What was learned about Epoxy Resin

UNITED INITIATORS MEKP INITIATOR – USAGE GUIDE

Initiator Concentration	Resin Quantity					
	Quart		Gallon		5 Gallons	
	Volume	Weight	Volume	Weight	Volume	Weight
1.00%	10 ml 0.3 fl oz	11 g 0.4 oz	40 ml 1.4 fl oz	44 g 1.5 oz	200 ml 6.8 fl oz	220 g 7.6 oz
1.25%	12.5 ml 0.4 fl oz	13.8 g 0.5 oz	50 ml 1.7 fl oz	55 g 1.9 oz	250 ml 8.5 fl oz	275 g 9.5 oz
1.50%	15 ml 0.5 fl oz	16.5 g 0.6 oz	60 ml 2.0 fl oz	66 g 2.3 oz	300 ml 10 fl oz	330 g 11.4 oz
1.75%	17.5 ml 0.6 fl oz	19.3 g 0.7 oz	70 ml 2.4 fl oz	77 g 2.7 oz	350 ml 12 fl oz	385 g 13 oz
2.00%	20 ml 0.7 fl oz	22 g 0.8 oz	80 ml 2.7 fl oz	88 g 3.0 oz	400 ml 13.5 fl oz	440 g 15 oz
2.50%	25 ml 0.9 fl oz	28 g 1.0 oz	100 ml 3.4 fl oz	110 g 3.8 oz	500 ml 17 fl oz	550 g 19 oz

ASSUMPTIONS: 1) Resin Weight 9.5 lbs/Gallon
2) MEKP specific gravity 1.1 grams/ml @ 77˚F (25˚C)
3) Initiator quantities rounded to the closest unit

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2^{nd} Stage Fin Manufacturing

- 2mm thick fiberglass
- Vacuum bag to remove air bubbles
- 3d printed mold

Control Surface Assembly

In House Manufactured

- Fin body (fiberglass)
- Fin Tab (3d printed)
- Shaft (1/4 in steel rod)

Purchased parts

- Servo to shaft couplers
- Bearings
- Screws

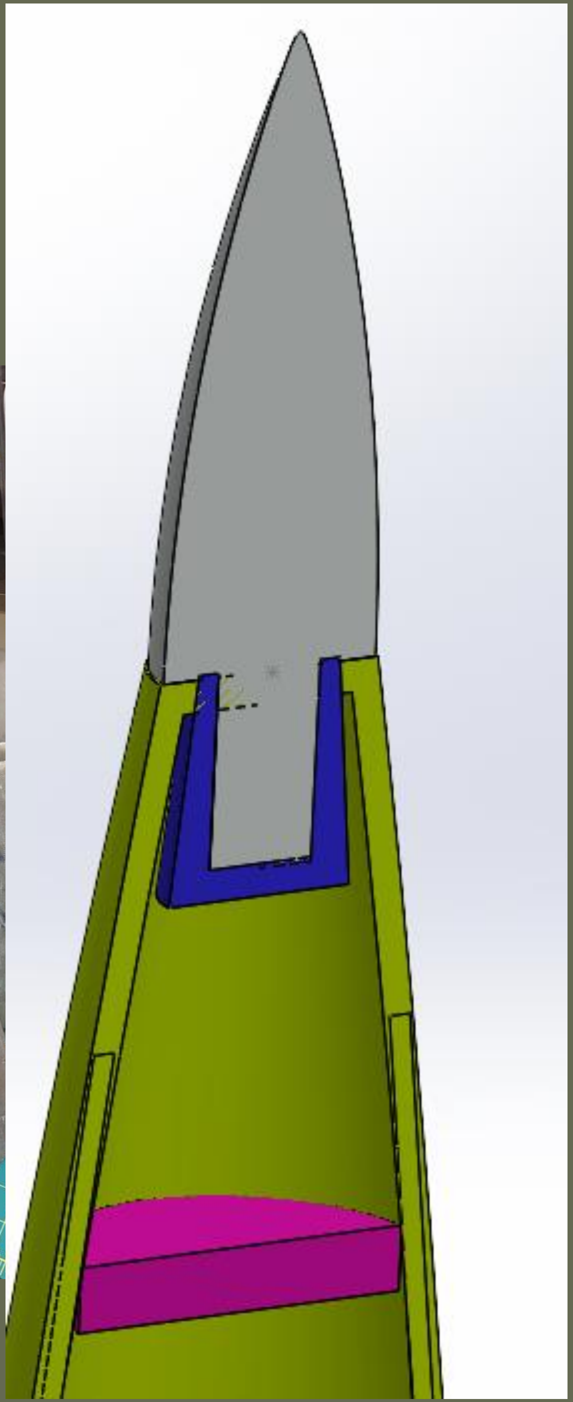
Control Surface Assembly

What didn’t work

- Servo to shaft couplers didn’t fit
- Solutions to this problem included plastic parts, and jb weld
- This was relatively unsuccessful

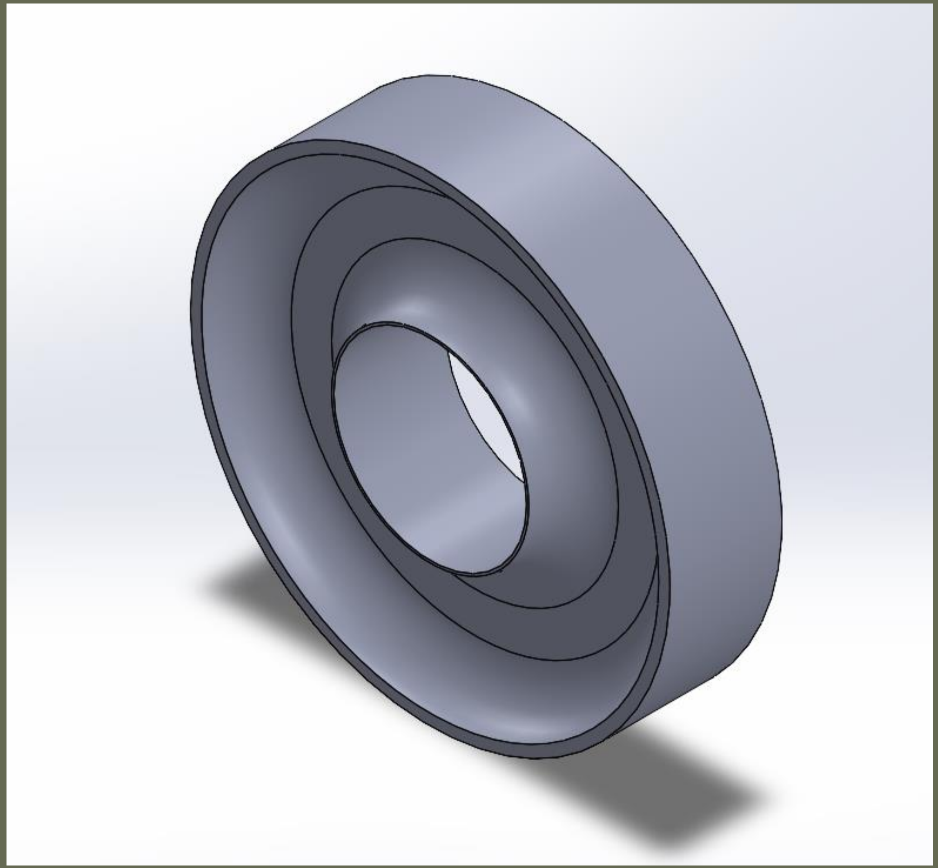
Nosecone Manufacturing

- 4mm thick fiberglass
- Vacuum bag to remove air bubbles
- 3d printed mold



Centering Ring Redesign

- Fillets had issues printing
- Increased top surface area
- Added skirts along the outside of the rings to increase surface adhesion

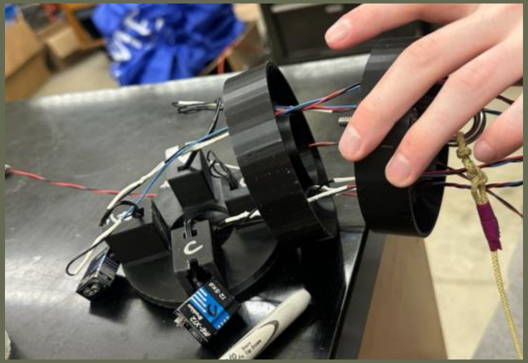


1st Stage Motor Mount Assembly

- Same as 2nd stage without wires

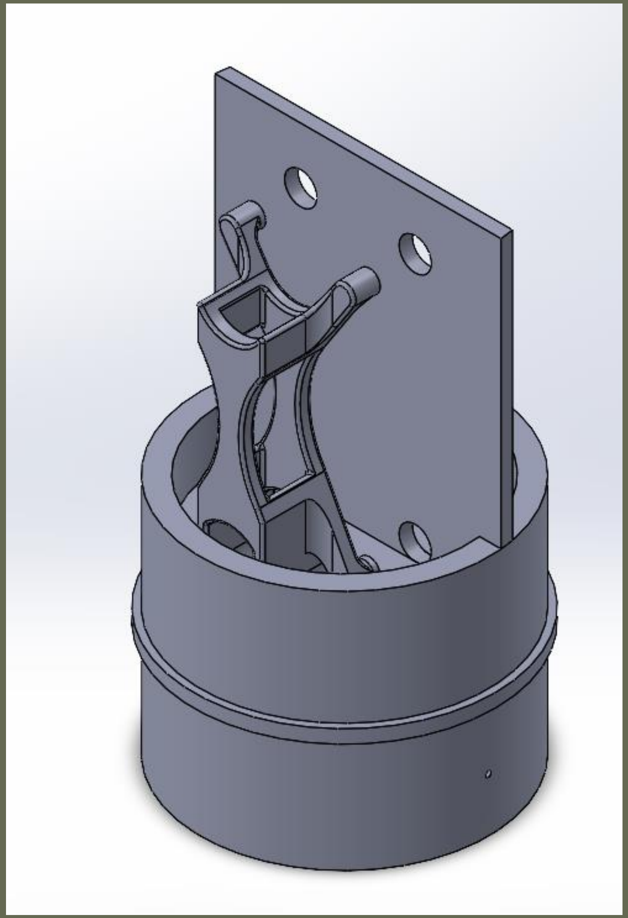
2nd Stage Motor Mount Assembly

- Assembled outside of body tube
- Epoxied top centering ring before motor tube
- Servos ran back through mounting ring before glueing in the bottom ring



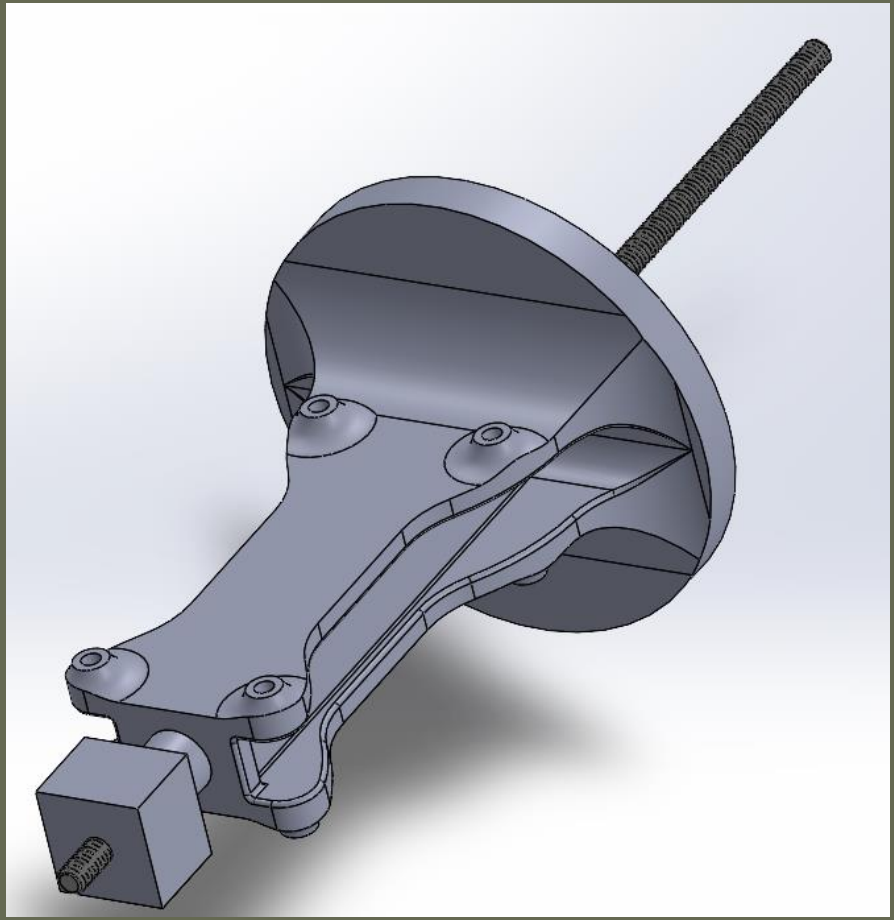
First Stage Electronics Bay

- Electronics
- Dual Deploy Bay
- Interstage Coupler

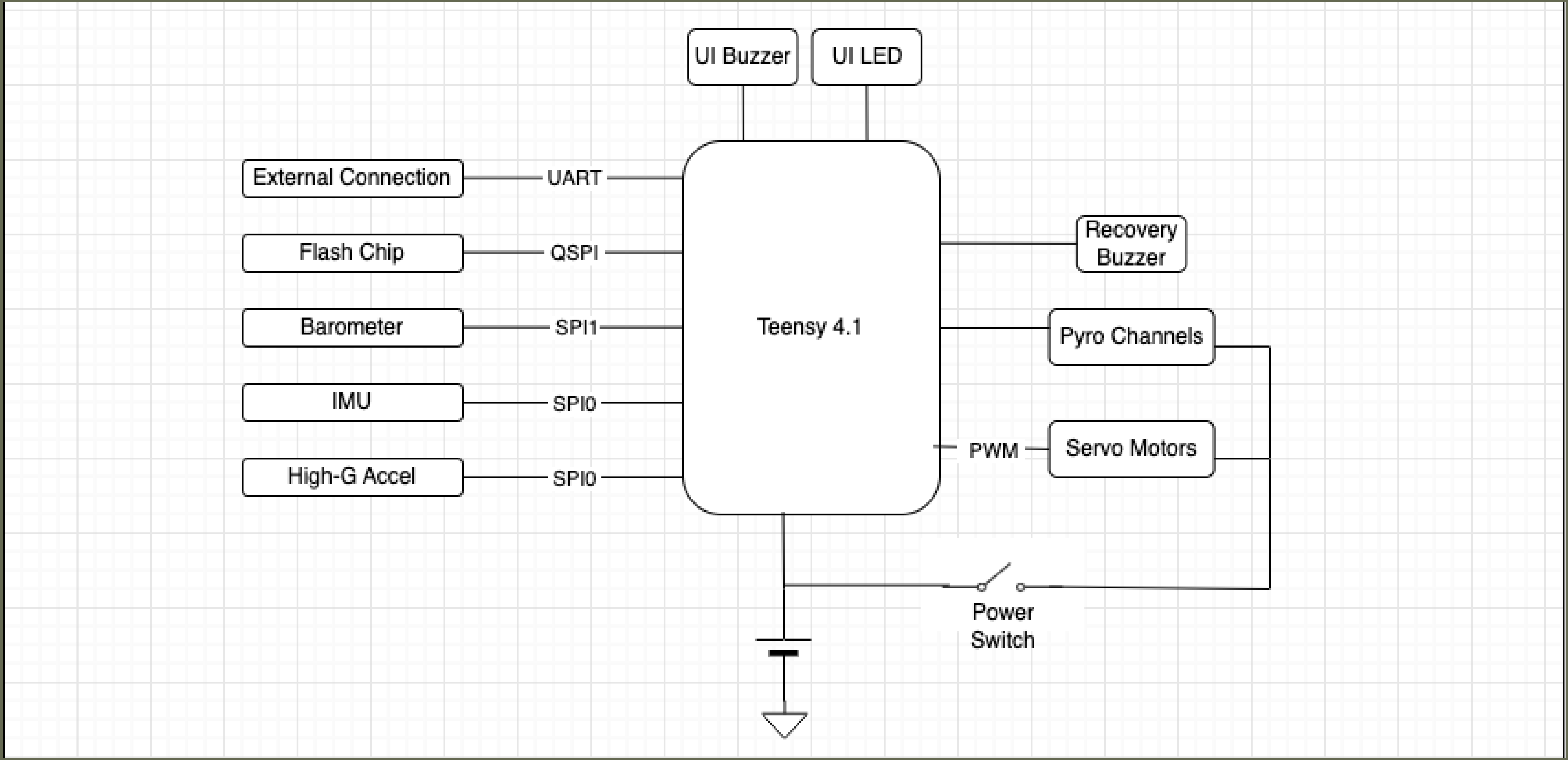


2nd Stage Electronics Bay Assembly

- Brass M3 Threaded Inserts
- Rubber washers to dampen vibration
- Large fillets for to reduce stress concentration
- Threaded rod to transfer force
- Space for 2 circuits boards, batteries



System Block Diagram



Sustainer Motor Safety Gates

- Power Switch Closed
- Software Arm
- External Connection Disconnected
- Pyro Channel Continuity Sensing
- Booster Coast Time Elapsed
- Tilt Lockout

Flight State Machines

Booster

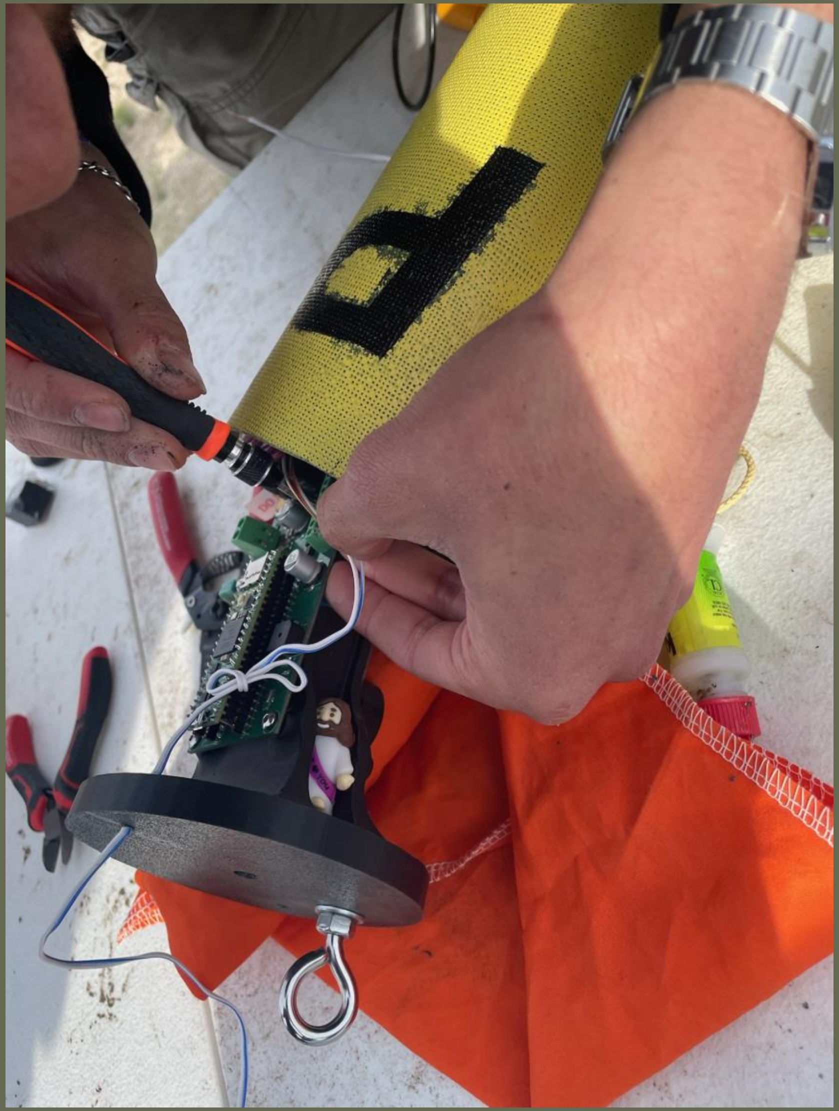
- Pre-Liftoff
- Booster Ignition
- Booster Coast
- Drogue Parachute
- Main Parachute
- Touchdown

Sustainer

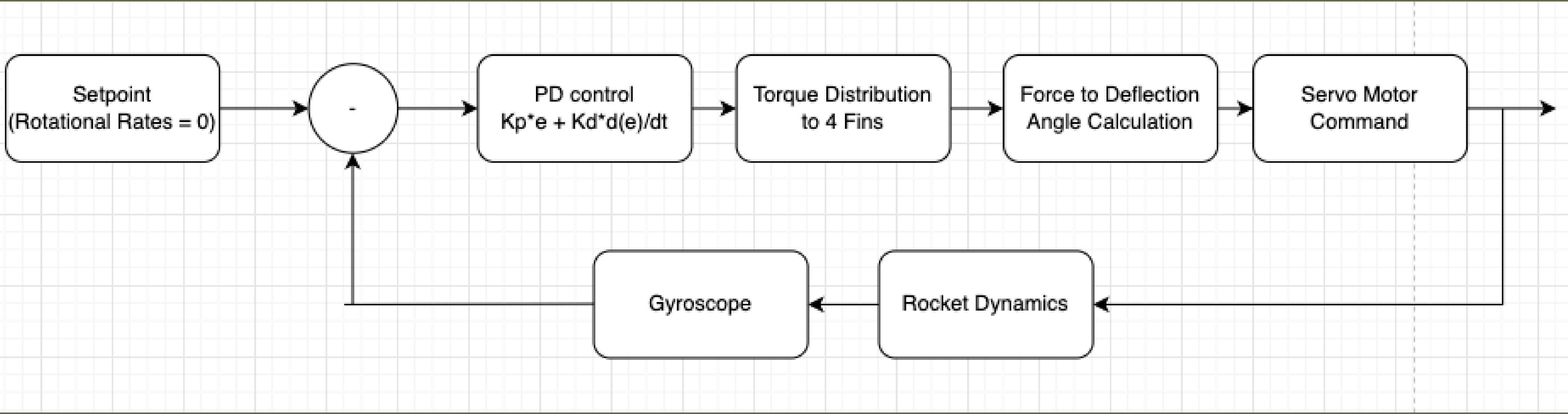
- Pre-Liftoff
- Booster Ignition
- Booster Coast
- Sustainer Ignition
- Sustainer Coast
- Drogue Parachute
- Main Parachute
- Touchdown

Pre-Flight Safety Protocol

- Command-Line Interface (CLI) over an External Connection
    - Calibrate Sensors
    - Software Arming
    - Battery Voltage Monitor
    - Check Servos
    - Pyro Channel Continuity
    - Check correct Firmware Upload



Control System Block Diagram

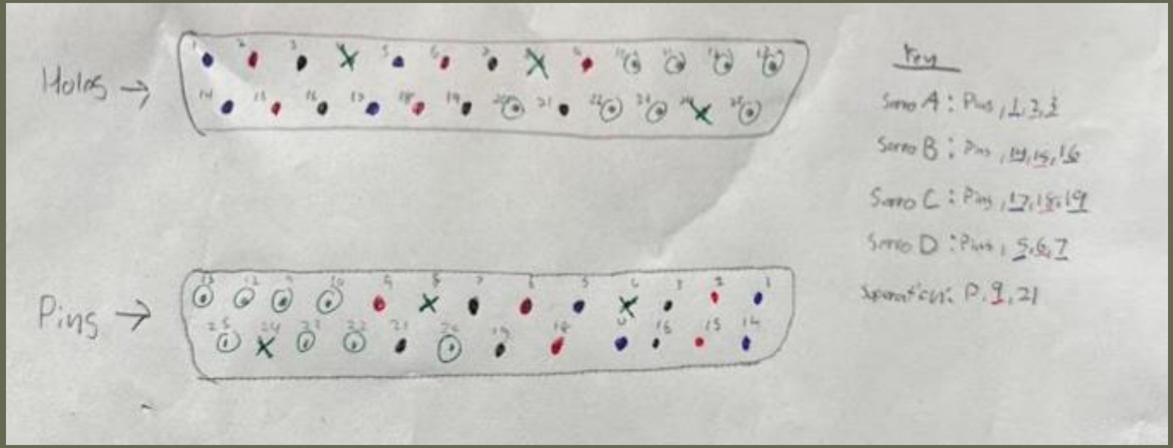


Servo Motor Misalignment

- During assembly, servo horns were pressed onto the splines at incorrect teeth
- Software trim/offset can correct residual misalignment to only a few degrees
- At commanded "neutral" (PWM 90°), some fin tabs sat 55°+ off parallel-to-airframe

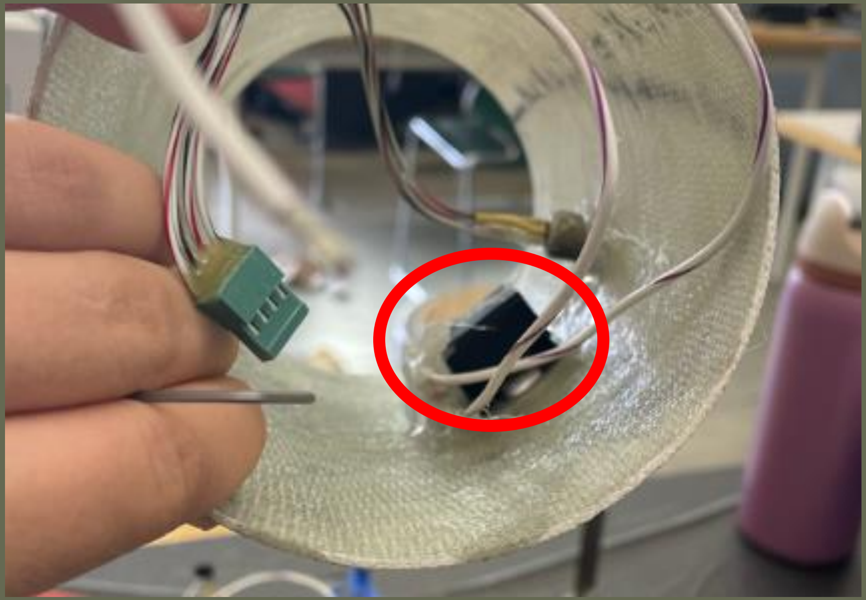
Sam Does Electrical Engineering

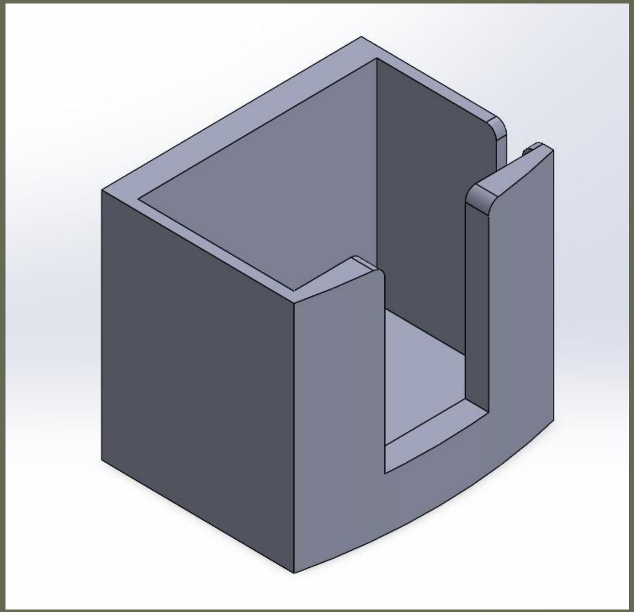
- Wires for servos and second stage ignition
- D25 used to separation electronics at apogee separation event
- Originally wanted to use VGA

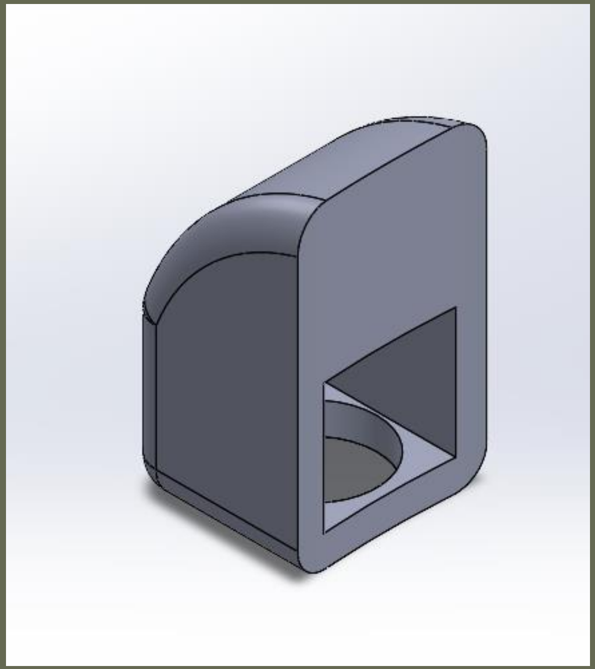


Camera Housing

- 3D Printed
- Broke off due to impact when installing e-bay at launch site







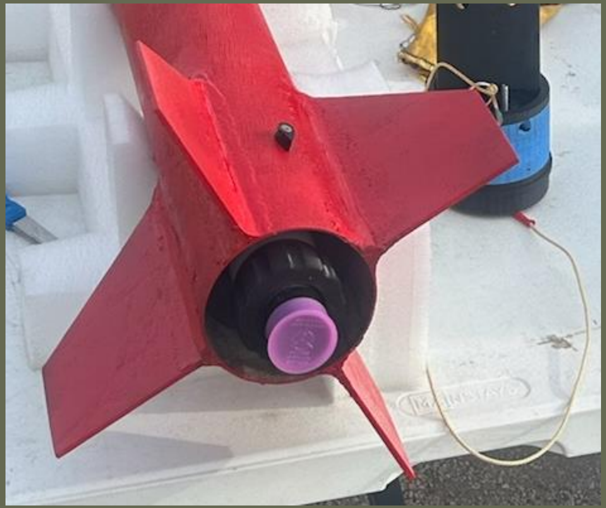
Motors

- L1000-18
  - Burn Time: 3s
  - Coast Time: 7s
  - Max Thrust: 1355N

- J435-14
  - Burn Time: 1.78s
  - Coast Time: 14s
  - Max Thrust: 696N

Motor Retention Caps

- Made to hold the motors in the mounting tubes
- 3d printed
- Epoxied onto the motor mount tubes



Recovery

- The recovery system went together as intended
- Assemblies approved for flight after intense questioning

Charges

- Initial
  - H335 Rifle Powder
  - Tested with 7g+
- Final
  - Triple Seven FFFG
  - 1g
  - Finer and more energetic than H335
  - Plastic Tubing
  - Hot glue
  - Paper towel

Charge Testing

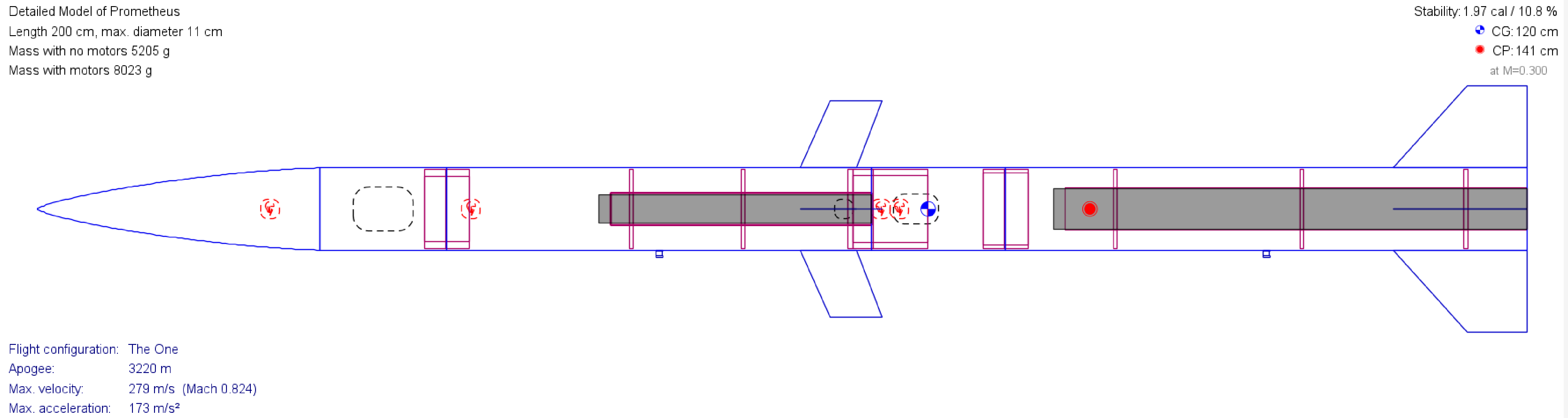
- H335 Rifle Powder Charges
  - Tested incrementally up to 7.5g
  - No tests separated the body tube from coupler with significant force
- Triple Seven FFFG Charges
  - Tested up to 1.5g
  - Tested with 0-2 shear pins
  - Settled on 1g

Final Assembly

- Chute lines tied off
- Parachutes packed
- Electronics bays aligned
- Shear pins installed
- Final checks made

OpenRocket Projections

- Max Velo: 279 m/s (Mach 0.8)
- Max Acceleration: 173 m/s^2
- Apogee: 3220 m (~10.5k ft)
- Weight: 8 kg
- Weight (no motors): 5.2 kg
- Real Weight: 4.92 kg

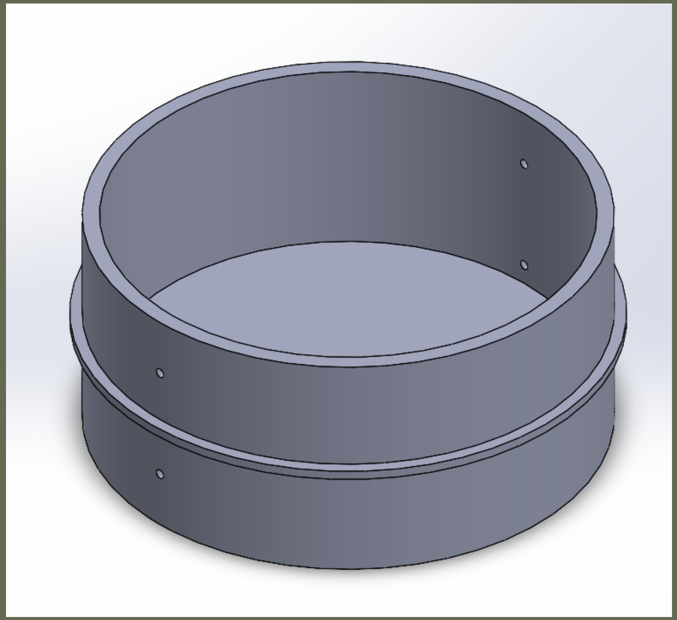


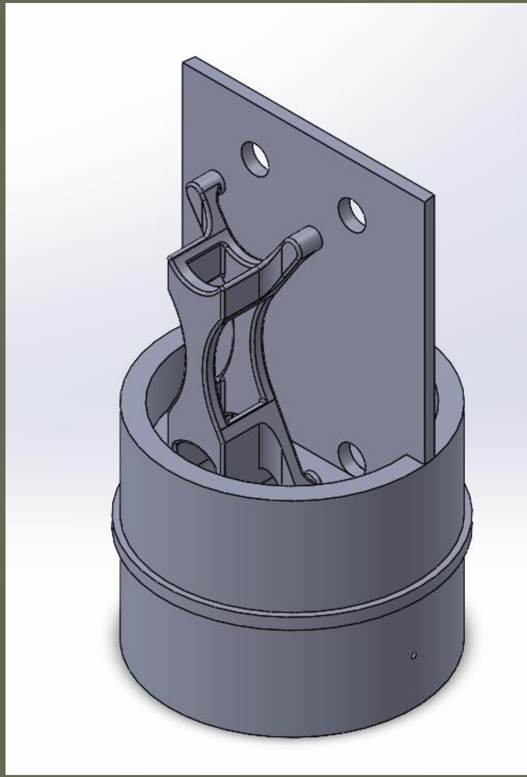
Budget

	Electronics	Motors	Miscellaneous	Total
Cost	$963.55	$442	$270.63	$1676.28

So Why Did Prometheus Not Launch?

- The couplers on Prometheus were deemed too short to be safely launched
- This was a colloquial rule that we were unaware of





Man Hours

- In total roughly 1400+ man hours to build
- Capstone requires 150 per person
- That's roughly 9 people in project hours

	Fall	Winter	Spring	Quarter Totals	Total
Dylan	64	87	124	275	1417
Sam	56	79	144	279	
Seth	65	71	144	280	
Jeremy	50	70	130	250	
Joseph	60	78	160	298	
Andrei	5	10	35	50	

Advice to Future Projects (That We Wish We Knew)

- Hardware-in-the-Loop (HIL) testing simulation
- Extensive materials testing on epoxy, PETG and assembly hardware

Problems from our Rocket:

- Redesign couplers
- Solve space issue
- Fix brass coupler(s)
- Materials testing
- Leave time for testing

Credits

- Andrei Vasilev
- Dr. Thomsen
- Reiff Manufacturing
- Dr. Ekkens
- Andrei Maiorov
- Kolby Kriegelstein
- Natalie Hunter
- Jess Mumek
- Dr. Ma
- Dr. Yaw
- Cristina Showalter
- Toby Pitton
- Carena Tomas
- Seth Waller
- David Bobocea
- Colten Hartman
- Xander Rusenescu
- ASME Club
- Sam's car
- Tree on Rogers field

Questions?

All uncredited photos were taken by the Prometheus rocket project team